DATASET DESCRIPTION

This document describes the structure of the dataset. There is a main folder named "Experimental Dataset of Charge and Discharge Patterns in a 1200 Ah OpzS Battery Bank."

Within the same folder are six ZIP-compressed subfolders (cell 1 through cell 6). After unzipping the ZIP files, each subfolder contains one corresponding to the data for one OPzS battery cell, containing 15 data files for cell 1 and 19 data files for cells 2 through 6, representing the different charge and discharge pattern tests performed.

Each file is named according to the following structure: "Cell measurements N – Test ordinal number – Charge pattern type – Discharge pattern type," where N represents the cell number. The nineteen files for each cell are listed below:

- Cell N Measurements - First Charge-C20 discharge.csv
- Cell N Measurements - Second Charge-C20 discharge.csv
- Cell N Measurements - Third Charge-C20 discharge.csv
- Cell N Measurements - Fourth Load-C20 discharge.csv
- Cell N Measurements - Fifth Load-C10 discharge.csv
- Cell N Measurements - Sixth Load-C10 discharge.csv
- Cell N Measurements - Seventh Photovoltaic Load clear - C20 discharge.csv
- Cell N Measurements - Seventh Photovoltaic Load Cloudy - C20 discharge.csv
- Cell N Measurements - Eighth Photovoltaic Load clear - C20 discharge.csv
- Cell N Measurements - Eighth Photovoltaic Load Cloudy - C20 discharge.csv
- Cell N Measurements - Ninth Photovoltaic Load clear - C10 discharge.csv
- Cell N Measurements - Ninth Photovoltaic Load Cloudy - C10 discharge.csv
- Cell N Measurements - Tenth Photovoltaic Load clear - C10 discharge.csv
- Cell N Measurements - Tenth Photovoltaic Load Cloudy - C10 discharge.csv
- Cell N Measurements - Eleventh Load - GC discharge.csv
- Cell N Measurements - Twelfth Photovoltaic Load Clear - GC discharge.csv
- Cell N Measurements - Twelfth Photovoltaic Load Cloudy - GC discharge.csv
- Cell N Measurements - Thirteenth Load wind energy - GC discharge.csv
- Cell N Measurements - Fourteenth Load - Discharge Pulses.csv

NOTE: In cell 1, the photovoltaic measurements on clear and cloudy days have been unified, which is why there are fewer files.

All files are provided in comma-separated values format (CSV extension). The delimiter used is a semicolon (;), which ensures clear separation of fields between columns. This format is widely compatible with data analysis software, databases, and programming languages such as MATLAB and Python, and facilitates data exchange between different applications.

Each file consists of a single main table containing ‘N’ records distributed across seven columns, organised in a clear tabular structure. In total, the dataset comprises approximately 540,000 individual records (around 90,000 per cell), distributed across the 14 experimental cycles.

The number of records per cycle varies considerably, from approximately 1,053 to 13,385 entries per cell (equivalent to between 6,318 and 80,310 records when considering the six cells), reflecting the different durations and operating conditions of the tests.

Therefore, the dataset is organised in a simple and consistent manner, facilitating efficient access, processing, and reuse in various analytical environments.

The columns are named according to the following table:

Column	Structural description	Format
Timestamp	Timestamp with one-minute resolution.	[DD/MM/YYYY hh:mm]
Vn (mV)	Voltage values in millivolts.	[XXXX,XX]
Vn (V)	Voltage values in volts (adjusted magnitude).	[X,XXXXX]
Tn (ºC)	Temperature values in degrees Celsius.	[XX,XX]
I (A)	Current in amperes (+ charge / - discharge).	[±XXX,XX]
(Ah)	Accumulated magnitude in ampere-hours.	[XXXX,XX]
%SOC	State of Charge percentage.	[XXX,XX]

NOTE: The value ‘n’ indicates the cell number.